Solve with us 6.2 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- Let $T: V \to W$T : V → W be a linear transformation. We define kernel of TT (denoted by $ker(T)$ ker(T)) and image of TT (denoted by $Im(T)$Im(T)) as follows:

$$Ker(T) = \{v \in V \mid T(v) = 0\}\text{Ker(T)} = \{\text{v} \in \text{V} \mid \text{T(v)} = 0\}$$
$$Im(T) = \{w \in W \mid \text{ there exists } v \in V \text{ for which } T(v) = w\}\text{Im(T)} = \{\text{w} \in \text{W} \mid \text{ there exists v} \in \text{V for which T(v) = w}\}$$

- TT is injective (i.e., TT is a monomorphism) if and only if $Ker(T) = \{0\}$Ker(T) = {0}.
- A linear transformation $T: V \to W$T : V → W is surjective (i.e., TT is epimorphism) if and only if $Im(T) = W$Im(T) = W.
- $Nullity(T) = dim(Ker(T))$Nullity(T) = dim(Ker(T)) and $Rank(T) = dim(Im(T))$ Rank(T) = dim(Im(T)).
- Let $T: V \to W$T : V → W be a linear transformation. Then

$$Rank(T) + Nullity(T) = dim(V).\text{Rank(T) + Nullity(T) = dim(V).}$$

This is known as the rank nullity theorem.

- Let $T: V \to W$T : V → W be a linear transformation. TT is injective if and only if $Nullity(T) = 0$ Nullity(T) = 0. TT is surjective if and only if $Rank(T) = dim(W)$Rank(T) = dim(W).

Consider the following linear transformation:

$$T: R^3 \to R^2 \qquad \text{T : R3 → R2}$$
$$T(x, y, z) = (2x + 3z, 4y + z) \qquad \text{T(x, y, z) = (2x + 3z, 4y + z)}$$

Answer the following questions:
1 point
Which of the following Options is true?

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$T(x, y, z) = (2x + 3z, 4y + z) = (0, 0), i.e., $2x + 3z = 0$ 2x + 3z = 0 and $4y + z = 0$4y + z = 0]

○
$ker(T) = \{(3y, y, -4y) \mid y \in R\}$ker(T) = {(3y, y, −4y) | y ∈ R}
○
$ker(T) = \{(6y, y, 4y) \mid y \in R\}$ker(T) = {(6y, y, 4y) | y ∈ R}
○
$ker(T) = \{(6y, y, -4y) \mid y \in R\}$ker(T) = {(6y, y, −4y) | y ∈ R}
○
$ker(T) = \{(3y, y, 4y) \mid y \in R\}$ker(T) = {(3y, y, 4y) | y ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$ker(T) = \{(6y, y, -4y) \mid y \in R\}$ker(T) = {(6y, y, −4y) | y ∈ R}
1 point
Which of the following is a basis of $Ker(T)$Ker(T)?
○

$\{(3, 1, -4)\}$

$\{(6, 0, -4), (0, 1, -4)\}$

$\{(6, 1, -4)\}$

$\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 1, -4)\}$

Find the nullity of T.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Find the rank of T.

[Hint: use rank nullity theorem.]

No, the answer is incorrect.

Score: 0

Feedback:

– Here dim(V) = 3 as V = R^3.

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Which of the following is true?

T is both injective and surjective.

T is injective but not surjective.

T is surjective but not injective.

T is neither injective nor surjective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is surjective but not injective.

1 point

Which option represents the kernel and image of the following linear transformation?

$$T: R^2 \rightarrow R^2$$
$$T(x, y) = (x, 0)$$

$ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$.

○
$ker(T) = Span\{(1,0)\}, Im(T) = Span\{(0,1)\}$ker(T) = Span{(1, 0)}, Im(T) = Span{(0, 1)}.
○
$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}.
○
$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(0,1)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(0, 1)}.

No, the answer is incorrect.
Score: 0
Feedback:
• Ker(T) = {(0, y) | y ∈ R}.
Accepted Answers:
$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}.

Key Points: 2Key Points: 2

- Let $T: V \rightarrow W$T : V → W be a linear. Follow the steps below to find bases for the null space and the range space of TT.

Step 1: Find the matrix AA corresponding to TT with respect to some standard ordered bases $\beta = \{v_1, v_2, ..., v_n\}$β = {v1, v2, ..., vn} and $\gamma = \{w_1, w_2, ..., w_m\}$γ = {w1, w2, ..., wm} for VV and WW respectively.
Step 2: Use row reduction on AA to obtain the matrix RR which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$Rx = 0 is the basis of null space of matrix AA and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \vdots \\ c_{2n} \end{bmatrix}, ..., \begin{bmatrix} c_{k1} \\ c_{k2} \\ \vdots \\ c_{kn} \end{bmatrix}$ $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ c11c12...c1n , $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ c21c22...c2n , ..., $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ ck1ck2...ckn form a

basis of the null space of AA precisely when the vectors $v_1', v_2', ..., v_k' \in$ ker(T)v1′, v2′, ..., vk′ ∈ ker(T) where $v_i' = \sum_{j=1}^{n} c_{ij} v_j$ vi′ = ∑j=1n cij vj, form a basis for ker(T)ker(T). Use the basis obtained in step 3 to thus get a basis for ker(T)ker(T).

Step 5: Recall that if $i_1, i_2, \ldots, i_r$ i1, i2, …, ir are the columns of RR containing the pivot elements, then the same columns of AA form a basis for the column space of AA.

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ \vdots \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \vdots \\ d_{2m} \end{bmatrix}, ..., \begin{bmatrix} d_{r1} \\ d_{r2} \\ \vdots \\ d_{rm} \end{bmatrix}$ $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ d11d12...d1m , $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ d21d22...d2m , ..., $\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$ dr1dr2...drm

form a basis of the column space of AA precisely when the vectors $w_1^{'}, w_2^{'}, ..., w_r^{'} \in im(T)$w1′, w2′, ..., wr′ ∈ im(T) where $w_i^{'} = \sum_{j=1}^{m} d_{ij} w_j$ wi′ = ∑j=1m dij wj, form a basis for $im(T)$im(T). Use the basis obtained in step 5 to thus get a basis for $im(T)$im(T).

Consider the following linear transformation:

$T: R^3 \rightarrow R^3$ T : R3 → R3
$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$ T(x, y, z) = (x − z, 2x + 3y + z, 3y + 3z)

Answer the following questions 7-9 using the information given above.
1 point
Which of the matrices corresponds the given linear transformation TT with respect to the standard ordered basis of R^3R3 for both the domain and the codomain?

○ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$

○ $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$

1 point
What will be the kernel of TT?

○ $\{(x, 0, z), (0, y, z) \mid x, y, z \in R\}$ {(x, 0, z), (0, y, z) | x, y, z ∈ R}.

○ $\{(z, 0, z), (0, -z, z) \mid z \in R\}$ {(z, 0, z), (0, −z, z) | z ∈ R}.

○ $\{(z, -z, z) \mid z \in R\}$ {(z, −z, z) | z ∈ R}.

○

$\{(x, 0, z), (0, -x, z) \mid x, z \in R\}$ $\{(x, 0, z), (0, -x, z) \mid x, z \in R\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(z, -z, z) \mid z \in R\}$ $\{(z, -z, z) \mid z \in R\}$.

1 point

What will be the basis of the image space of T T found by the algorithm mentioned in the Key Points ?

○

$\{(1, 0, -1), (2, 3, 1)\}$ $\{(1, 0, -1), (2, 3, 1)\}$

○

$\{(1, 0, 0), (0, 1, 0)\}$ $\{(1, 0, 0), (0, 1, 0)\}$

○

$\{(1, 2, 0), (0, 3, 3)\}$ $\{(1, 2, 0), (0, 3, 3)\}$

○

$\{(-1, 1, 3)\}$ $\{(-1, 1, 3)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 2, 0), (0, 3, 3)\}$ $\{(1, 2, 0), (0, 3, 3)\}$

Check Answers

Your score is: 0/9